POST-ROUTE MULTI-CORNER STATIC TIMING ANALYSIS OF VSDBABYSOC USING OPENROAD AND SKY130HD

1. INTRODUCTION

Static Timing Analysis (STA) is an essential technique used to verify the timing performance of digital integrated circuits without applying simulation test vectors. It evaluates signal propagation delays and determines whether data signals meet the required setup and hold timing constraints.

The timing performance of a circuit varies with process, voltage, and temperature (PVT) conditions. Multi-corner STA analyzes a design using different timing libraries representing these operating conditions. This helps identify critical timing paths and potential timing violations.

In this work, post-route STA is performed on the VSDBabySoC design using OpenROAD and SKY130HD standard-cell timing libraries. The routed design database, timing constraints, and extracted parasitics are used to analyze setup and hold timing across 16 PVT corners.

2. AIM

To perform post-route Static Timing Analysis of the VSDBabySoC design across 16 SKY130HD timing corners using OpenROAD and evaluate timing performance using slack, Worst Negative Slack (WNS), and Total Negative Slack (TNS).

3.Theory

The analysis uses 16 SKY130HD timing views covering fast-fast (FF), slow-slow (SS), and typical-typical (TT) process conditions at different temperatures and supply voltages.

The following timing parameters are considered:

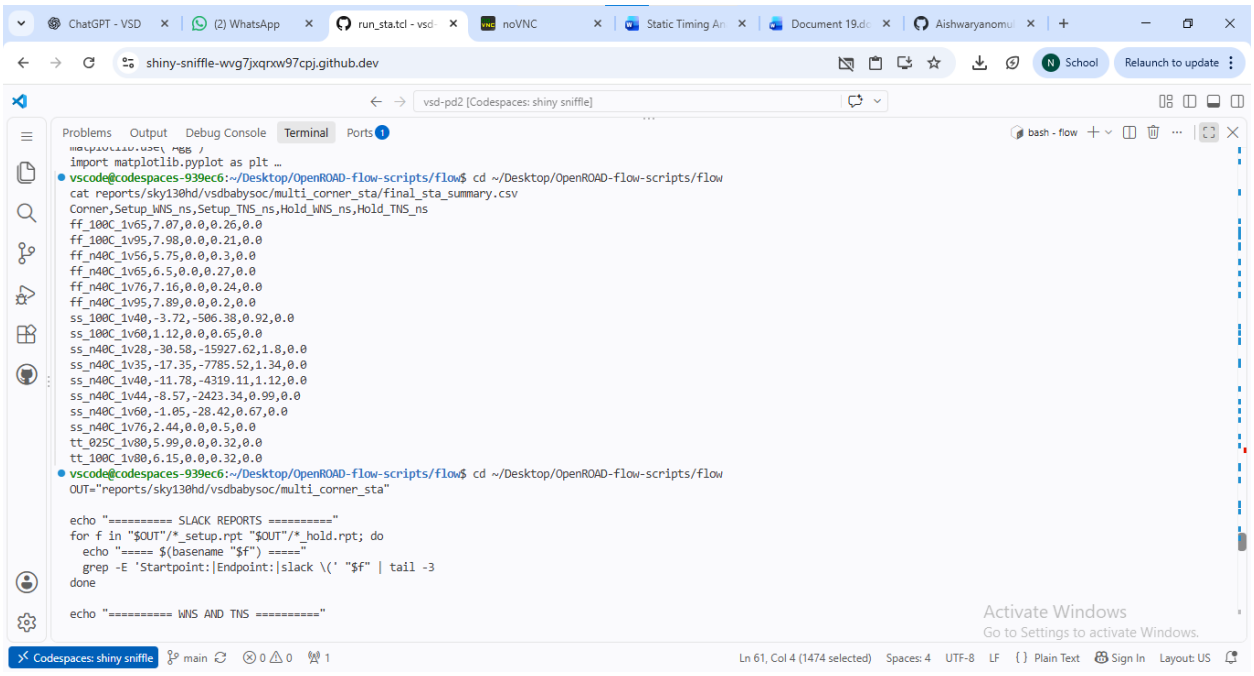
- **Slack:** The difference between the required arrival time and the actual arrival time of a signal. Negative setup slack indicates a timing violation.
- **Worst Negative Slack (WNS):** The worst slack value among the analyzed timing paths. A negative setup WNS indicates that at least one setup path fails its timing requirement.
- **Total Negative Slack (TNS):** The sum of negative slack values across violating timing paths. A TNS of zero indicates that no negative slack is accumulated for the reported timing check.

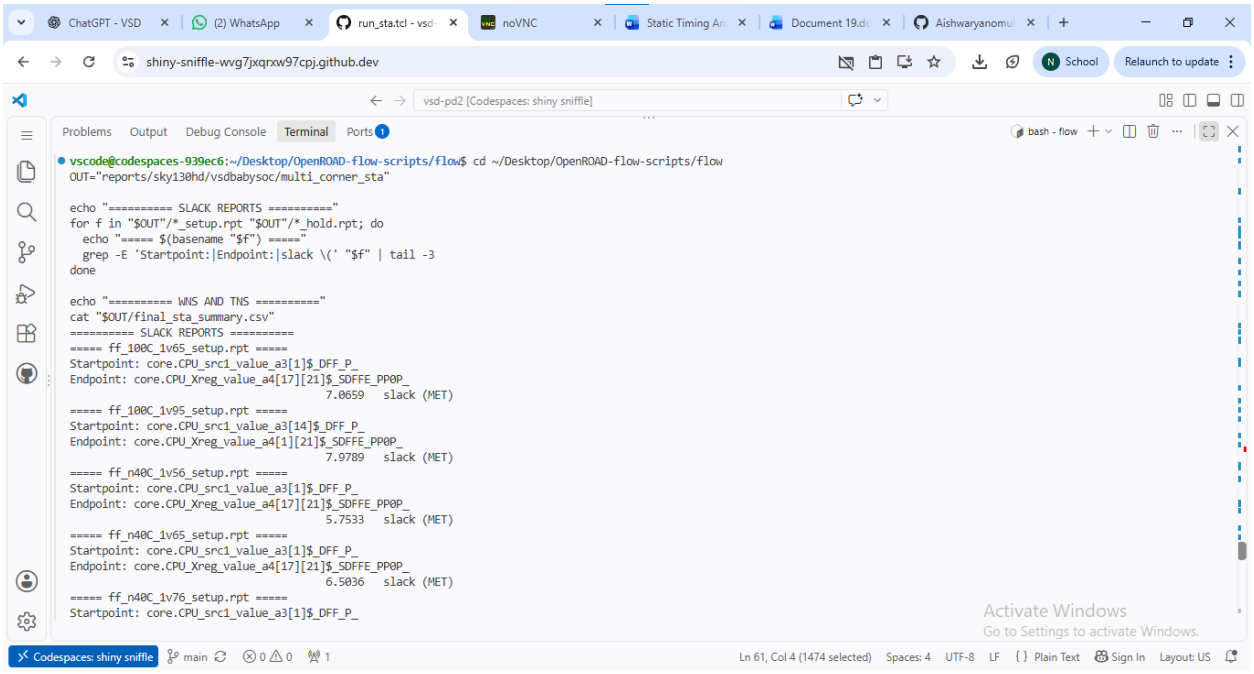
Setup analysis checks whether data arrives before the required clock edge. Hold analysis checks whether data remains stable for the required time after the clock edge.

4. TOOLS AND TECHNOLOGIES

1. **OpenROAD:** Used to load the routed design and perform static timing analysis.
2. **VSDBabySoC:** The digital SoC design under analysis.
3. **SKY130HD timing libraries:** Used to represent the 16 PVT timing corners.

5. RESULTS AND DISCUSSION

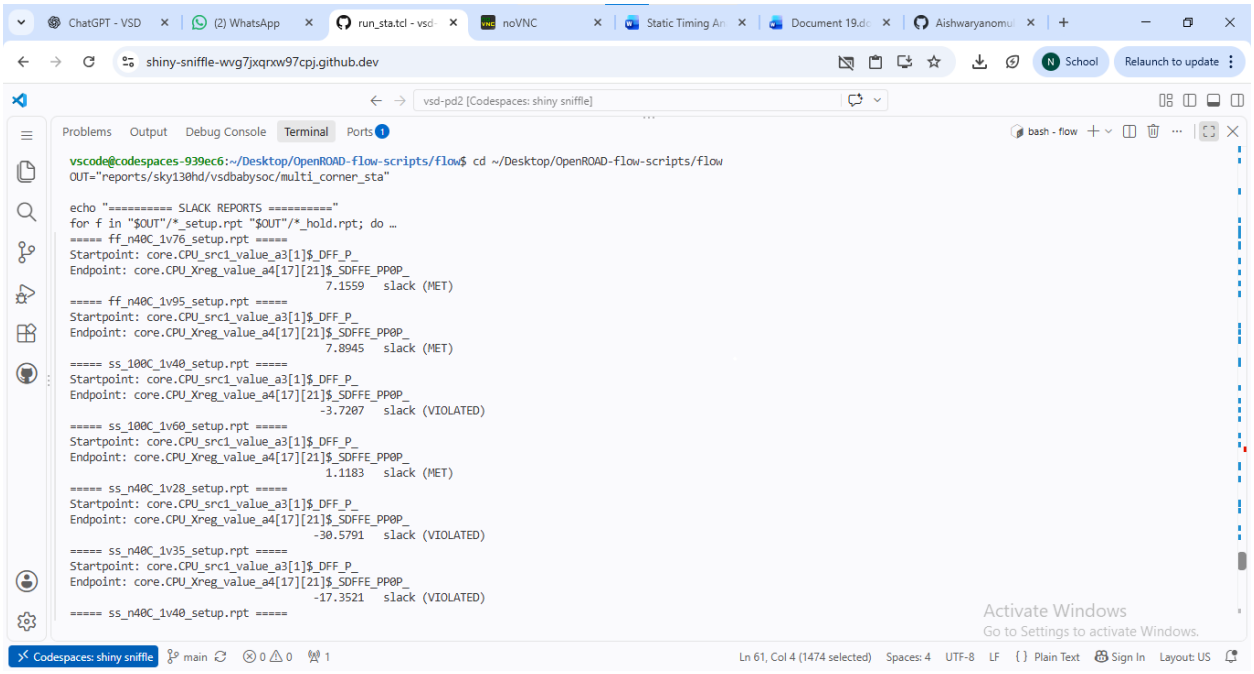




```
vscode@codespaces-939ec6:~/Desktop/OpenROAD-flow-scripts/flow$ cd ~/Desktop/OpenROAD-flow-scripts/flow
OUT="reports/sky130hd/vsdbabysoc/multi_corner_sta"

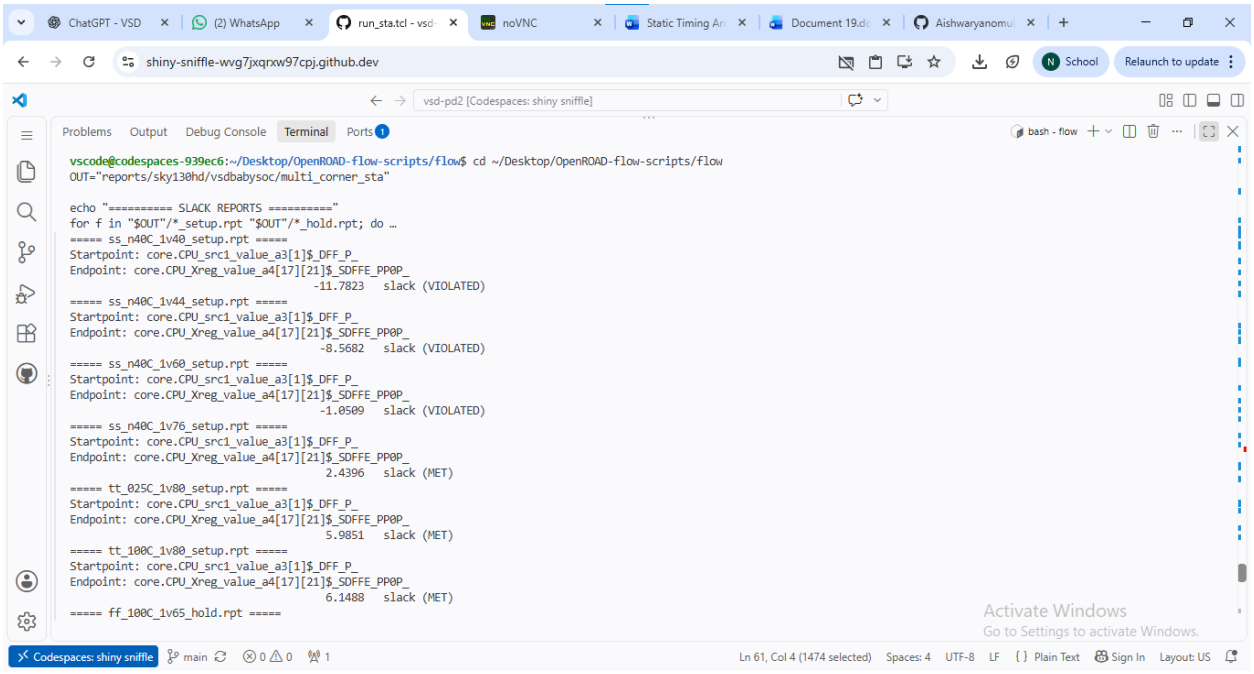
echo "========== SLACK REPORTS =========="
for f in "$OUT"/*_setup.rpt "$OUT"/*_hold.rpt; do
  echo "===== $(basename "$f") ====="
  grep -E 'Startpoint:|Endpoint:|slack \(' "$f" | tail -3
done

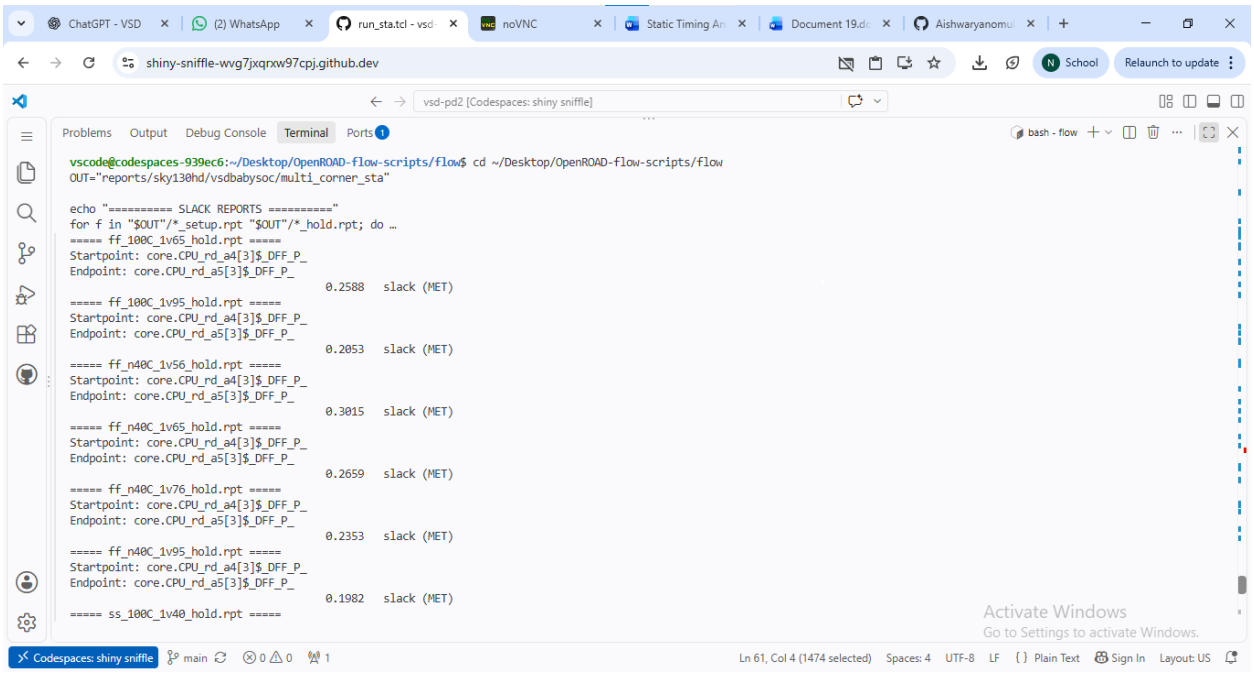
echo "========== WNS AND TNS =========="
cat "$OUT/final_sta_summary.csv"
========== SLACK REPORTS ==========
===== ff_100C_1v65_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                  7.0659   slack (MET)
===== ff_100C_1v95_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[14]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[1][21]$_SDFFE_PP0P_
                                  7.9789   slack (MET)
===== ff_n40C_1v56_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                  5.7533   slack (MET)
===== ff_n40C_1v65_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                  6.5036   slack (MET)
===== ff_n40C_1v76_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
```

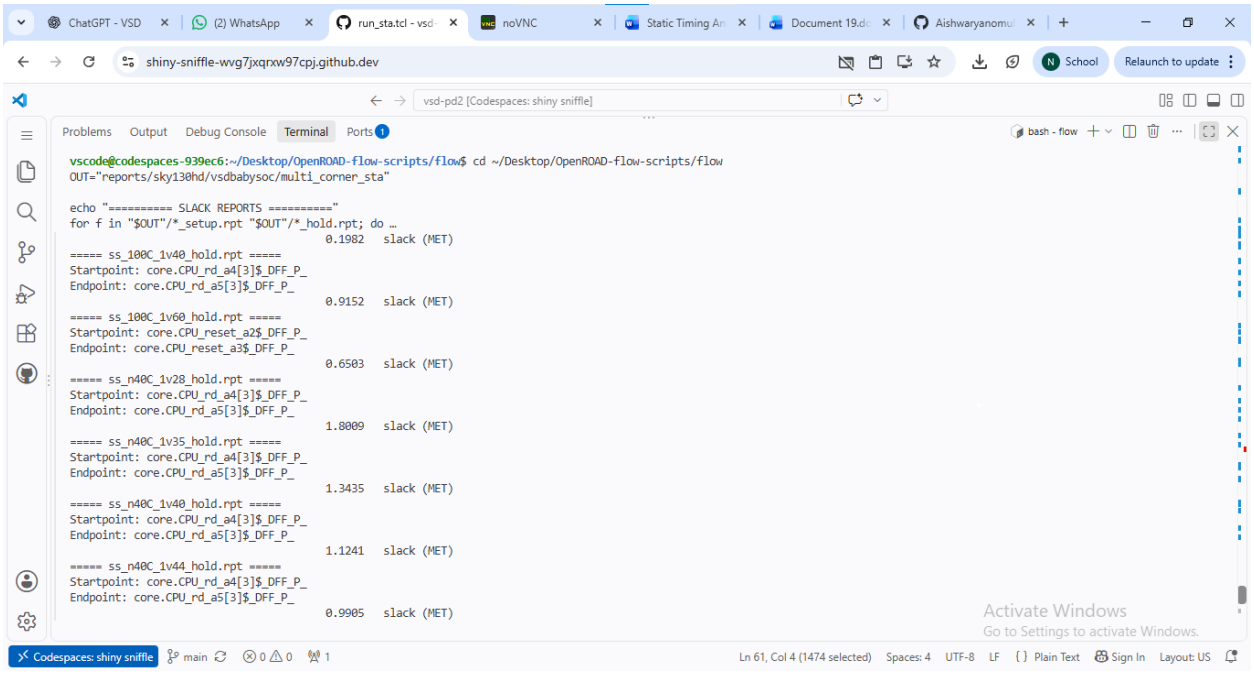


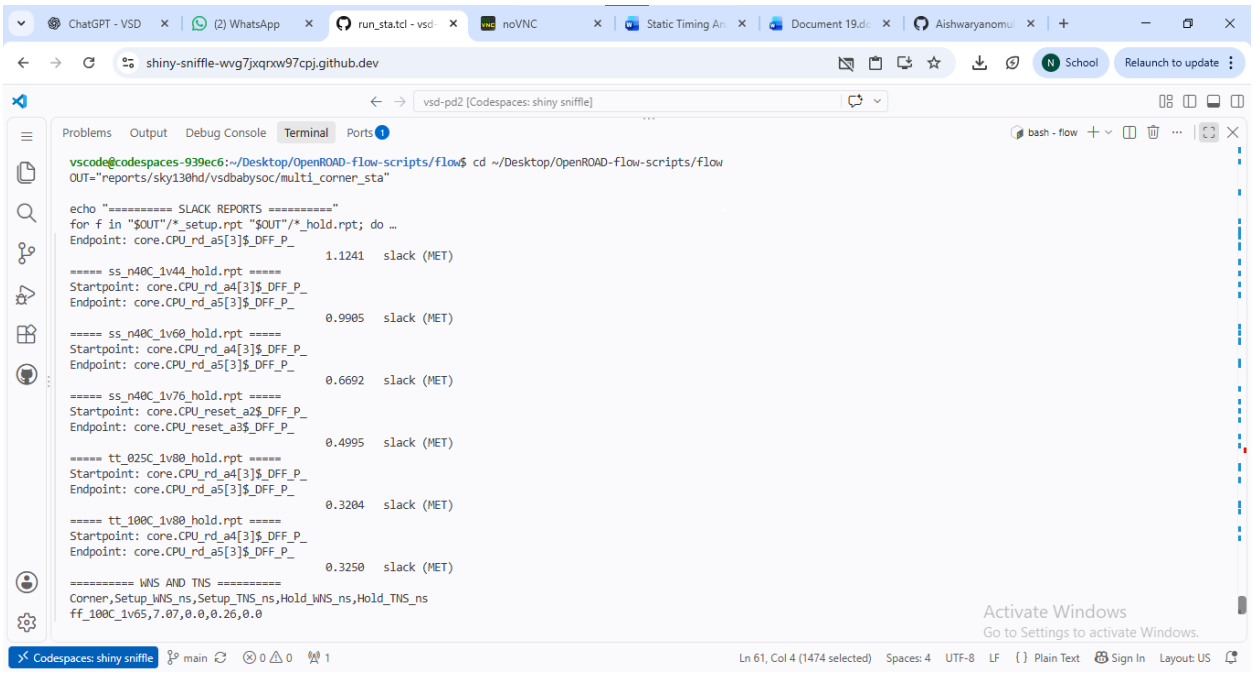
```
vscode@codespaces-939ec6:~/Desktop/OpenROAD-flow-scripts/flow$ cd ~/Desktop/OpenROAD-flow-scripts/flow
OUT="reports/sky130hd/vsdbabysoc/multi_corner_sta"

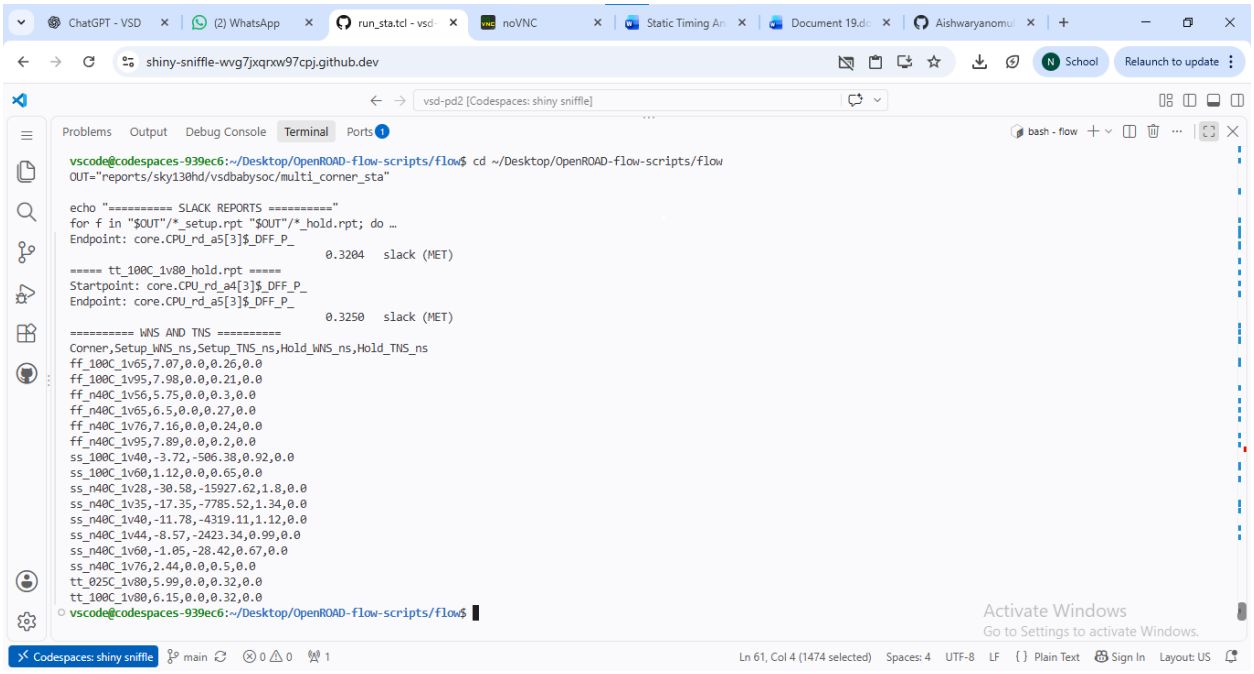
echo "========== SLACK REPORTS =========="
for f in "$OUT"/*_setup.rpt "$OUT"/*_hold.rpt; do …
===== ff_n40C_1v76_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                  7.1559   slack (MET)
===== ff_n40C_1v95_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                  7.8945   slack (MET)
===== ss_100C_1v40_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                 -3.7207   slack (VIOLATED)
===== ss_100C_1v60_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                  1.1183   slack (MET)
===== ss_n40C_1v28_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                -30.5791   slack (VIOLATED)
===== ss_n40C_1v35_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
                                -17.3521   slack (VIOLATED)
===== ss_n40C_1v40_setup.rpt =====
```


shiny-sniffle-wvg7jxqrxw97cpj.github.dev
vsd-pd2 [Codespaces: shiny sniffle]
Problems Output Debug Console Terminal Ports
vscode@codespaces-939ec6:~/Desktop/OpenROAD-flow-scripts/flow$ cd ~/Desktop/OpenROAD-flow-scripts/flow
OUT="reports/sky130hd/vsdbabysoc/multi_corner_sta"
echo "========== SLACK REPORTS =========="
for f in "$OUT"/*_setup.rpt "$OUT"/*_hold.rpt; do …
===== ss_n40C_1v40_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
-11.7823 slack (VIOLATED)
===== ss_n40C_1v44_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
-8.5682 slack (VIOLATED)
===== ss_n40C_1v60_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
-1.0509 slack (VIOLATED)
===== ss_n40C_1v76_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
2.4396 slack (MET)
===== tt_025C_1v80_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
5.9851 slack (MET)
===== tt_100C_1v80_setup.rpt =====
Startpoint: core.CPU_src1_value_a3[1]$_DFF_P_
Endpoint: core.CPU_Xreg_value_a4[17][21]$_SDFFE_PP0P_
6.1488 slack (MET)
===== ff_100C_1v65_hold.rpt =====
Activate Windows
Go to Settings to activate Windows.
Codespaces: shiny sniffle
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shiny-sniffle-wvg7jxqrxw97cpj.github.dev
vsd-pd2 [Codespaces: shiny sniffle]
Problems Output Debug Console Terminal Ports
vscode@codespaces-939ec6:~/Desktop/OpenROAD-flow-scripts/flow$ cd ~/Desktop/OpenROAD-flow-scripts/flow
OUT="reports/sky130hd/vsdbabysoc/multi_corner_sta"
echo "========== SLACK REPORTS =========="
for f in "$OUT"/*_setup.rpt "$OUT"/*_hold.rpt; do …
===== ff_100C_1v65_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.2588 slack (MET)
===== ff_100C_1v95_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.2053 slack (MET)
===== ff_n40C_1v56_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.3015 slack (MET)
===== ff_n40C_1v65_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.2659 slack (MET)
===== ff_n40C_1v76_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.2353 slack (MET)
===== ff_n40C_1v95_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.1982 slack (MET)
===== ss_100C_1v40_hold.rpt =====
Activate Windows
Go to Settings to activate Windows.
Codespaces: shiny sniffle
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shiny-sniffle-wvg7jxqrxw97cpj.github.dev
Problems Output Debug Console Terminal Ports
vscode@codespaces-939ec6:~/Desktop/OpenROAD-flow-scripts/flow$ cd ~/Desktop/OpenROAD-flow-scripts/flow
OUT="reports/sky130hd/vsdbabysoc/multi_corner_sta"
echo "========== SLACK REPORTS =========="
for f in "$OUT"/*_setup.rpt "$OUT"/*_hold.rpt; do …
0.1982 slack (MET)
===== ss_100C_1v40_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.9152 slack (MET)
===== ss_100C_1v60_hold.rpt =====
Startpoint: core.CPU_reset_a2$_DFF_P_
Endpoint: core.CPU_reset_a3$_DFF_P_
0.6503 slack (MET)
===== ss_n40C_1v28_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
1.8009 slack (MET)
===== ss_n40C_1v35_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
1.3435 slack (MET)
===== ss_n40C_1v40_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
1.1241 slack (MET)
===== ss_n40C_1v44_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.9905 slack (MET)
Activate Windows
Go to Settings to activate Windows.
Codespaces: shiny sniffle
Ln 61, Col 4 (1474 selected) Spaces: 4 UTF-8 LF Plain Text Sign In Layout: US


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Problems Output Debug Console Terminal Ports
vscode@codespaces-939ec6:~/Desktop/OpenROAD-flow-scripts/flow$ cd ~/Desktop/OpenROAD-flow-scripts/flow
OUT="reports/sky130hd/vsdbabysoc/multi_corner_sta"
echo "========== SLACK REPORTS =========="
for f in "$OUT"/*_setup.rpt "$OUT"/*_hold.rpt; do …
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
1.1241 slack (MET)
===== ss_n40C_1v44_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.9905 slack (MET)
===== ss_n40C_1v60_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.6692 slack (MET)
===== ss_n40C_1v76_hold.rpt =====
Startpoint: core.CPU_reset_a2$_DFF_P_
Endpoint: core.CPU_reset_a3$_DFF_P_
0.4995 slack (MET)
===== tt_025C_1v80_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.3204 slack (MET)
===== tt_100C_1v80_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
0.3250 slack (MET)
========== WNS AND TNS ==========
Corner,Setup_WNS_ns,Setup_TNS_ns,Hold_WNS_ns,Hold_TNS_ns
ff_100C_1v65,7.07,0.0,0.26,0.0
Activate Windows
Go to Settings to activate Windows.
Codespaces: shiny sniffle
Ln 61, Col 4 (1474 selected) Spaces: 4 UTF-8 LF Plain Text Sign In Layout: US



```
vscode@codespaces-939ec6:~/Desktop/OpenROAD-flow-scripts/flow$ cd ~/Desktop/OpenROAD-flow-scripts/flow
OUT="reports/sky130hd/vsdbabysoc/multi_corner_sta"

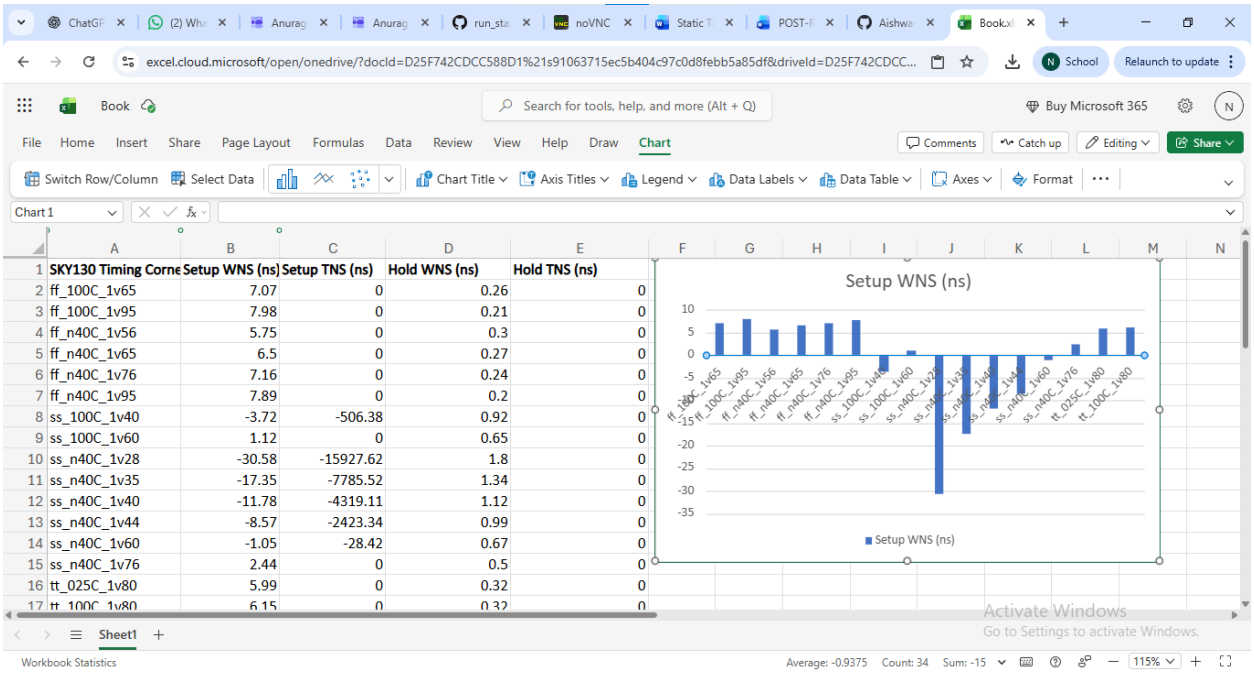
echo "========== SLACK REPORTS =========="
for f in "$OUT"/*_setup.rpt "$OUT"/*_hold.rpt; do …
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
                                    0.3204   slack (MET)

===== tt_100C_1v80_hold.rpt =====
Startpoint: core.CPU_rd_a4[3]$_DFF_P_
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
                                    0.3250   slack (MET)

========== WNS AND TNS ==========
Corner,Setup_WNS_ns,Setup_TNS_ns,Hold_WNS_ns,Hold_TNS_ns
ff_100C_1v65,7.07,0.0,0.26,0.0
ff_100C_1v95,7.98,0.0,0.21,0.0
ff_n40C_1v56,5.75,0.0,0.3,0.0
ff_n40C_1v65,6.5,0.0,0.27,0.0
ff_n40C_1v76,7.16,0.0,0.24,0.0
ff_n40C_1v95,7.89,0.0,0.2,0.0
ss_100C_1v40,-3.72,-506.38,0.92,0.0
ss_100C_1v60,1.12,0.0,0.65,0.0
ss_n40C_1v28,-30.58,-15927.62,1.8,0.0
ss_n40C_1v35,-17.35,-7785.52,1.34,0.0
ss_n40C_1v40,-11.78,-4319.11,1.12,0.0
ss_n40C_1v44,-8.57,-2423.34,0.99,0.0
ss_n40C_1v60,-1.05,-28.42,0.67,0.0
ss_n40C_1v76,2.44,0.0,0.5,0.0
tt_025C_1v80,5.99,0.0,0.32,0.0
tt_100C_1v80,6.15,0.0,0.32,0.0
vscode@codespaces-939ec6:~/Desktop/OpenROAD-flow-scripts/flow$
```

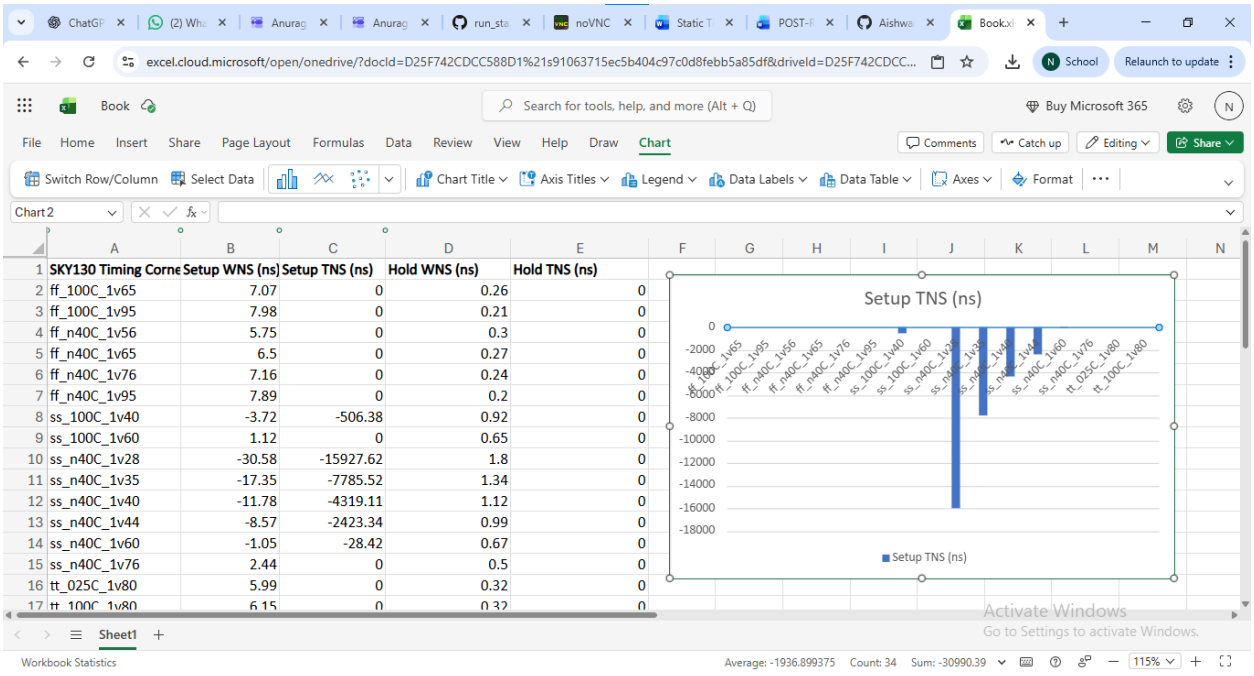
	A	B	C	D	E	F	G	H	I
1	SKY130 Timing Corne	Setup WNS (ns)	Setup TNS (ns)	Hold WNS (ns)	Hold TNS (ns)				
2	ff_100C_1v65	7.07	0	0.26	0				
3	ff_100C_1v95	7.98	0	0.21	0				
4	ff_n40C_1v56	5.75	0	0.3	0				
5	ff_n40C_1v65	6.5	0	0.27	0				
6	ff_n40C_1v76	7.16	0	0.24	0				
7	ff_n40C_1v95	7.89	0	0.2	0				
8	ss_100C_1v40	-3.72	-506.38	0.92	0				
9	ss_100C_1v60	1.12	0	0.65	0				
10	ss_n40C_1v28	-30.58	-15927.62	1.8	0				
11	ss_n40C_1v35	-17.35	-7785.52	1.34	0				
12	ss_n40C_1v40	-11.78	-4319.11	1.12	0				
13	ss_n40C_1v44	-8.57	-2423.34	0.99	0				
14	ss_n40C_1v60	-1.05	-28.42	0.67	0				
15	ss_n40C_1v76	2.44	0	0.5	0				
16	tt_025C_1v80	5.99	0	0.32	0				
17	tt_100C_1v80	6.15	0	0.32	0				
18									
19									

WNS:



SKY130 Timing Corne	Setup WNS (ns)	Setup TNS (ns)	Hold WNS (ns)	Hold TNS (ns)
ff_100C_1v65	7.07	0	0.26	0
ff_100C_1v95	7.98	0	0.21	0
ff_n40C_1v56	5.75	0	0.3	0
ff_n40C_1v65	6.5	0	0.27	0
ff_n40C_1v76	7.16	0	0.24	0
ff_n40C_1v95	7.89	0	0.2	0
ss_100C_1v40	-3.72	-506.38	0.92	0
ss_100C_1v60	1.12	0	0.65	0
ss_n40C_1v28	-30.58	-15927.62	1.8	0
ss_n40C_1v35	-17.35	-7785.52	1.34	0
ss_n40C_1v40	-11.78	-4319.11	1.12	0
ss_n40C_1v44	-8.57	-2423.34	0.99	0
ss_n40C_1v60	-1.05	-28.42	0.67	0
ss_n40C_1v76	2.44	0	0.5	0
tt_025C_1v80	5.99	0	0.32	0
tt_100C_1v80	6.15	0	0.32	0

TNS:



SKY130 Timing Corne	Setup WNS (ns)	Setup TNS (ns)	Hold WNS (ns)	Hold TNS (ns)
ff_100C_1v65	7.07	0	0.26	0
ff_100C_1v95	7.98	0	0.21	0
ff_n40C_1v56	5.75	0	0.3	0
ff_n40C_1v65	6.5	0	0.27	0
ff_n40C_1v76	7.16	0	0.24	0
ff_n40C_1v95	7.89	0	0.2	0
ss_100C_1v40	-3.72	-506.38	0.92	0
ss_100C_1v60	1.12	0	0.65	0
ss_n40C_1v28	-30.58	-15927.62	1.8	0
ss_n40C_1v35	-17.35	-7785.52	1.34	0
ss_n40C_1v40	-11.78	-4319.11	1.12	0
ss_n40C_1v44	-8.57	-2423.34	0.99	0
ss_n40C_1v60	-1.05	-28.42	0.67	0
ss_n40C_1v76	2.44	0	0.5	0
tt_025C_1v80	5.99	0	0.32	0
tt_100C_1v80	6.15	0	0.32	0